

Benjamin Pomeranz

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EDUCATION

Brown University

Providence, RI

Sc.B. in Mathematics and A.B. in Computer Science, 4.0 GPA

Sep. 2022 – May 2026

- **Math:** Graduate real function theory, manifolds, real and complex analysis, number theory, cryptography, linear and abstract algebra, graph theory, differential geometry, modern applications of probability, honors statistics
- **Computer Science:** Deep Learning (Implementing various neural nets in TF and Pytorch), Graduate Algorithms (Linear Programming, Graph Sparsification, Multiplicative Weights), Cryptography, Computer Systems, etc.
- **Budapest Semester in Mathematics:** Awarded highest honors at intensive study abroad program in math; courses in Galois theory, Benjamini-Schramm convergence of graphs, algebraic topology, and “Conjecture and Proof”. Researched the 3 dimensional 2 player subtraction game.
- Courses in nonfiction writing

EMPLOYMENT AND RESEARCH

Pivotal Fellowship with SecureBio

London, UK

Research fellow

June 2026 – August 2026

- Using item response theory to improve the “Biological Capabilities Index”
- Creating internal tooling to audit the information content and difficulty distribution of benchmarks

REU in Computational Mathematics at Montana State University

Bozeman, MT

Researcher in metabolic models

June 2025 – August 2025

- Developed and implemented algorithm in Python to 2x speed of minimal EFM decomposition for flux vectors and enable numerical stability on genome scale models for first time
- Leveraged convex geometry of the flux cone and linear programming to prove efficacy of algorithm
- Collected and analyzed data on algorithmic performance using Pandas, NumPy, and SciPy
- Presented research at JMM, and currently completing transcript to be published with Dr. Bree Cummins

Aqua Voice, Summer Intern

San Francisco, CA

Intern

June – August 2024

- Implemented new windowing algorithm in Aqua’s Python backend, improving latency by $\sim 3x$
- Automated model finetuning and evaluation processes to reduce dictation errors by $1.5x$
- Integrated an open source voice activity detection model, improving accuracy and latency

Brown University CS Department

Providence, RI

TA for CS200: Data Structures and Algorithms I

January-May 2024

- Teaching students arithmetic computation, sorting, BSTs, hash tables, graph algorithms, and DP
- Running labs and hours with up to 30 students

Brown University Athletics Department, Student Athlete Tutor

Providence, RI

Tutored collegiate student athletes at Brown University

October – December 2023

- Tutored in Math 0100 (Calculus II) and Math 0540 (Linear Algebra with theory)

ORGANIZATIONAL EXPERIENCE

Brown Effective Altruism

Providence, RI

Copresident

June 2025 – Present

- Leading intro fellowship reading group on existential risks, global health, and animal welfare
- Helping organize and designing website and data management for EA Northeast Retreat 2026
- Organizing monthly speaker presentations, monthly socials, and joint events with other clubs

Tenaflly High School

Tenaflly, NJ

Student Organization President

Sep. 2021 – June 2022

SKILLS AND MISCELLANY

Languages: Python, SQL **Rusty Languages:** Java, C

General Productivity: LaTeX, Google Suite, Linear, Notion, written communication, public speaking

Hobbies: Running, hiking, climbing, ultimate frisbee, crosswords, reading